

Year-over-Year Sales Growth

Using Python and Power BI DAX

Teacher: Nigussie Guluma — Data Analysis Lesson

Learning Goal

Students will understand how to calculate, interpret, and compare **Year-over-Year (YoY) Sales Growth** using two tools: **Python pandas** and **Power BI DAX**.

1. What is Year-over-Year Sales Growth?

Year-over-Year Sales Growth compares sales from one year to the sales from the previous year. It helps answer this question:

“Compared to last year, did sales increase or decrease?”

Formula

$$\text{YoY Growth \%} = \frac{\text{Current Year Sales} - \text{Previous Year Sales}}{\text{Previous Year Sales}} \times 100$$

Interpretation:

- Positive percentage means sales increased.
- Negative percentage means sales decreased.
- Zero percent means sales stayed the same.

2. Sample Sales Data

Year	Sales	Meaning
2021	\$40,000	Starting year
2022	\$46,000	Sales increased
2023	\$52,900	Sales increased again
2024	\$50,000	Sales decreased
2025	\$65,000	Sales increased strongly

3. Manual Example

Find the YoY growth from 2024 to 2025.

$$\frac{65000 - 50000}{50000} \times 100$$

$$\frac{15000}{50000} \times 100 = 30\%$$

Conclusion

From 2024 to 2025, sales increased by **30%**. This means the company sold much more in 2025 than in 2024.

4. Python Method Using pandas

Python is useful when students want to calculate sales growth from a table or dataset. The pandas function `pct_change()` calculates percent change from one row to the next row.

Python Code

```
import pandas as pd
import matplotlib.pyplot as plt

# Step 1: Create the dataset
data = {
    "Year": [2021, 2022, 2023, 2024, 2025],
    "Sales": [40000, 46000, 52900, 50000, 65000]
}

df = pd.DataFrame(data)

# Step 2: Calculate previous year sales
df["Previous Year Sales"] = df["Sales"].shift(1)

# Step 3: Calculate YoY growth amount
df["YoY Growth"] = df["Sales"] - df["Previous Year Sales"]

# Step 4: Calculate YoY growth percent
df["YoY Growth %"] = df["Sales"].pct_change() * 100

print(df)
```

Expected Python Output

Year	Sales	Previous Year Sales	YoY Growth	YoY Growth %
2021	40,000	—	—	—
2022	46,000	40,000	6,000	15.00%
2023	52,900	46,000	6,900	15.00%
2024	50,000	52,900	-2,900	-5.48%
2025	65,000	50,000	15,000	30.00%

Python Graph Code

```
# Bar graph for total sales by year
plt.figure(figsize=(8, 5))
plt.bar(df["Year"], df["Sales"])
plt.title("Total Sales by Year")
plt.xlabel("Year")
```

```
plt.ylabel("Sales")
plt.show()

# Line graph for YoY Growth Percentage
plt.figure(figsize=(8, 5))
plt.plot(df["Year"], df["YoY Growth %"], marker="o")
plt.title("Year-over-Year Sales Growth %")
plt.xlabel("Year")
plt.ylabel("YoY Growth %")
plt.axhline(0, linestyle="--")
plt.show()
```

Python Explanation

`shift(1)` moves the sales values down one row so each year can be compared with the previous year.
`pct_change()` calculates the percentage change automatically.

5. Power BI Method Using DAX

Power BI is useful when students want to build dashboards and interactive reports. DAX measures calculate values dynamically based on filters, slicers, and visuals.

Data Model Requirement

For YoY DAX calculations to work correctly, use a proper Date Table.

Table	Column
Sales	Sales[SalesAmount]
Sales	Sales[Date]
Calendar	Calendar[Date]

Create a relationship:

Calendar[Date] → Sales[Date]

Create a Date Table in DAX

```
Calendar =
CALENDAR(
    MIN(Sales[Date]),
    MAX(Sales[Date])
)
```

Add useful columns:

```
Year = YEAR(Calendar[Date])
Month = FORMAT(Calendar[Date], "MMMM")
Month Number = MONTH(Calendar[Date])
```

DAX Measures

Measure 1: Total Sales

```
Total Sales =
SUM(Sales[SalesAmount])
```

Measure 2: Previous Year Sales

```
Sales Previous Year =
CALCULATE(
    [Total Sales],
    SAMEPERIODLASTYEAR(Calendar[Date])
)
```

Measure 3: YoY Sales Growth Amount

```
YoY Sales Growth =
[Total Sales] - [Sales Previous Year]
```

Measure 4: YoY Sales Growth Percentage

```

YoY Sales Growth % =
DIVIDE(
    [YoY Sales Growth],
    [Sales Previous Year],
    0
)

```

Format **YoY Sales Growth %** as a percentage in Power BI.

6. Python vs. Power BI Comparison

Question	Python	Power BI
Best use	Data analysis and coding practice	Dashboard and business reporting
Main tool	pandas	DAX measures
Formula method	<code>pct_change()</code> or manual formula	<code>SAMEPERIODLASTYEAR()</code> and <code>DIVIDE()</code>
Output	DataFrame, graphs, calculations	Interactive tables, charts, slicers
Good for students because	They see how the calculation is built step by step	They learn business intelligence reporting

7. Student Practice

Use the following data to answer the questions.

Year	Sales
2020	\$35,000
2021	\$40,000
2022	\$46,000
2023	\$52,900
2024	\$50,000
2025	\$65,000

1. Calculate the YoY growth amount from 2020 to 2021.
2. Calculate the YoY growth percentage from 2021 to 2022.
3. Which year had negative growth?
4. Which year had the highest growth percentage?
5. In Python, which function can calculate percent change automatically?
6. In DAX, which function returns sales from the previous year?
7. Explain in one sentence why YoY growth is useful for a business.

8. Exit Ticket

1. What does YoY stand for?
2. Write the formula for YoY Growth %.
3. If sales increased from \$80,000 to \$100,000, what is the YoY Growth %?
4. Which tool is better for dashboards: Python or Power BI?
5. Which tool is better for writing code and practicing data analysis: Python or Power BI?

Teacher Note

Both Python and Power BI use the same mathematical idea. Python helps students understand the calculation through code. Power BI helps students communicate results through dashboards and business reports.